Answers on
Booklet - 1

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

10^{-9} .

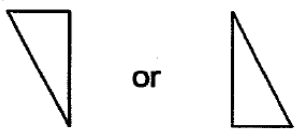
c .....

mega

tera .

2.

(a)(i)	vector triangle drawn (with arrows) showing the correct directions	B1
	suitable scale / correct working	B1
	velocity 256 (kmh^{-1}) / 71.1 m s^{-1}	A1
	direction 21° (west of north) (20.5) / 339	A1
	if scale diagram used:	
	(allow velocity of 240 to 270 and angle of 19 to 22)	



(ii) **1**	40 (km)	A1
2	15 (km)	A1

TOTAL [6]

3.

Question 1			Expected Answers	Marks	Additional Guidance
(a)			A vector <u>has a direction</u>	B1	Ignore additional comments unless there is a contradiction
(b)			Kinetic energy, mass and power all correct (no others underlined)	B1	
(c)	(i)	1	Tcos20 = W (W = 650) T = 692 (N) allow 2 sf	C1 A1	Solution involving triangles: correct triangle and three arrows C1 Scale diagram requires scale and correct triangle (no arrows)C1 allow +/- of 20 (N) for scale diagram do not allow 691 (N) but penalise only once on the paper
		2	Tsin20 = R R = 692sin20 = 237 236(.6) (N) Allow 240 (N)	A1	Allow ecf for T from (c)(i)1 Allow +/- 220 to 250 (N) for scale diagram Correct answer for 1 and 2 **in reverse** can score 1 mark
	(ii)		With the angle made with rock face less the tension is closer to equalling the weight value or example quoted cos10 is greater than cos20 i.e. closer to one	A1	Expect clear answer here. Do not allow answers such as R decreases therefore T increases. Allow answers that refer to cos angle is greater (for smaller angles) hence T will be less
			Total	6	

4. **(a)** energy and power S **B1**
velocity and displacement V **B1**
(if more than two answers are given then –1 for each one incorrect. Minimum score is zero)

(b)(i) average speed = 200 / 24 **C1**
= 8.3 (m s^{-1}) **A1**

(ii) velocity is less/speed is greater **B1**
displacement (SF) is less than distance (SF) **B1**

TOTAL [6]

5.

(a)(i) speed is distance per unit time / distance / time B1

(ii) velocity is the displacement per unit time / displacement / time B1

(iii) speed is a scalar, velocity is a vector B1
velocity has a direction B1
(speed has magnitude, velocity had magnitude and direction scores 2)

(b)(i) **1** speed = (40 x π) / (105 x 2) C1
= 0.60 (m s^{-1}) allow 1 s.f. A1

2 velocity = 40 / 105 C1
= 0.38 (m s^{-1}) A1

(ii) **1** 0.60 (m s^{-1}) allow ecf ((b)(i)) A1

2 0 A1
suitable comment for 2 e.g. there is no displacement B1

(c) different magnitudes B1
different directions B1
directions:
the average velocity is along the diameter / upwards B1
the instantaneous velocity is tangent to the circle / to the left B1
magnitudes:
instantaneous velocity equals average speed B1
average velocity is less as displacement is less B1

MAX 3

TOTAL [14]

6.

(a)	scalar has magnitude/ size/ value	B1
	vector has direction and magnitude/ size/ value	B1
	(vector has direction and scalar does not: scores ½)	
(b)	velocity	B1
	momentum	B1
	acceleration	B1
	(-1 for each error and extra quantity given)	
(c)(i)	6 cos30 = 5.2 (N)	B1
(ii)	6 sin30 = 3(.0) (N) possible ECF	B1
	(if answers are left not calculated then –1 if answers reversed then –1)	

7.

(a)	rate of change of velocity / change in velocity per unit time	A1
(b)(i)	correct velocity read from graph (18)	C1
	acceleration = (18 – 0) / 10 = 1.8 (m s^{-2})	A1
(ii)	area under the graph (maybe implied from calculations)	C1
	(treated as two triangles and a rectangle) areas = 90 + 2160 + 180	C1
	distance = 2430 m	A1

8.

distance = area under line or $(v/2)t$ or $s = (8.8)^2 / (2 \times 9.81)$
$= 8.8 / 2 \times 0.90 = 3.96\,\text{m}$ or $s = 3.95\,\text{m} = 4(.0)\,\text{m}$

acceleration $= (-4.4 - 8.8) / 0.50$
$= (-)\ 26(.4)\,\text{m s}^{-2}$

the accelerations are constant as straight lines

the accelerations are the same as same gradient or
no air resistance as acceleration is constant or
change of speed in opposite directions (one speeds up one slows down)

9.

(a) (i)	1. Acceleration $= 0 - 25 / 60$	C1
	$= -0.42$ (m s^{-2}) (one mark for correct sign)	A2
	(Do not allow 0.4 or 0.41 (ms^{-2})	
	2. distance is area under graph / = 25 x 60 / 2	C1
	= 750 (m)	A1
(ii)	distance is area under graph / = 25 x 40 / 2	C1
	= 500 (m)	A1
(iii)	time taken stopping at station = 60 + 90 + 40	
	= 190 (s)	C1
	distance travelled 1250 (m)	C1
	time taken by non stopping train 1250/25 = 50 (s)	C1
	time lost = 190 – 50	
	= 140 (s)	A1
	alternatives	
	time taken 230/ 240/ 280	C1
	distance travelled 2250/ 2500/ 3500	C1
	time take 90/ 100/ 140	C1

10.

(a)			**Gradient equated to acceleration, either explicitly or via a correct example** **PLUS ANY <u>TWO</u> FROM:** **Constant acceleration / deceleration** **<u>OR</u> Gradient negative, so negative acceleration / deceleration** **<u>At 2.0</u> s direction of velocity reverses / begins to move the other way** **OR Same speeds / at 0 and 4 s / at start and end**	**B1** **B1** **B1** **MAX 3**	 Allow only one mark for 'acceleration' comment and one mark for 'velocity' comment.